10 Devices, 0.2 probability.

P of 0 Transmission $= (0.8)^{10}$

P of 1 Transmission $= 10 \times 0.2 \times (0.8)^9$

For no collision we can add these both

$P(\text{no collision}) = P(0) + P(1)$

$= 0.1074 + 0.2684$

$= 0.3758$

$C = 8$ - - - - - - - - -

$a-z = 26$, $A-Z = 26$, $0-9 = 10$

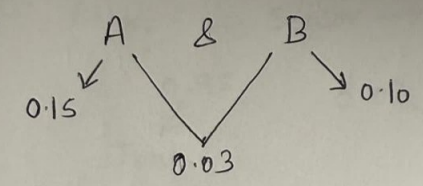
We can find at least one digit with finding opposi

$P(\text{no digit}) = \dfrac{P(\text{characters})}{\text{Total}} = \left(\dfrac{52}{62}\right)^8$

$\therefore P(\text{at least one digit}) = 1 - \left(\dfrac{52}{62}\right)^8$

$= 0.755$

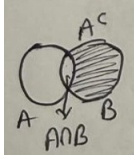
3.



$P(A) = 0.15$, $P(B) = 0.10$, $A(A \cap B) = 0.03$

(a) $A(A|B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.03}{0.10} = 0.30$

(b) $P(B|A^c) = \dfrac{P(B \cap A^c)}{P(A^c)} = \dfrac{P(B) - P(A \cap B)}{P(A^c)}$



$\boxed{P(A^c) = 1 - 0.15 = 0.85}$

$= \dfrac{0.10 - 0.03}{0.85} = 0.0823$

4. Robot $\rightarrow$ Sensor $\rightarrow$ Obstacle

$0.95 \swarrow$ True

$0.02 \searrow$ False Detection

Obstacles — 20% of the time

<u>Using Bayes Theorem</u>:-

P(O) for obstacle

P(S) for sensor

$$P(O|S) = \frac{P(O)P(S|O)}{P(S)}$$

$$P(S) = P(S|O)P(O) + P(S|O^c)P(O^c)$$
$$= 0.113$$

$$P(O|S) = \frac{0.95 \times 0.10}{0.113}$$
$$= 0.841$$

5. $P(S) = 0.10$

$P(M|S) = 0.80$

$P(M|S^c) = 0.05$

Bayes, $P(S|M) = \dfrac{P(S)P(M|S)}{P(M)}$

$$P(M) = P(M|S)P(S) + P(M|S^c)P(S^c)$$
$$= (0.80 \times 0.10) + (0.05 \times 0.90)$$
$$= 0.125$$

$$P(S|M) = \frac{0.80 \times 0.10}{0.125} = 0.64$$

6. $P(D) = 1 \text{ in } 10{,}000 = 0.0001$

$P(T|D) = 0.99$

$P(T|D^c) = 0.005$

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)}$$

$$P(T) = P(T|D)P(D) + P(T|D^c)P(D^c)$$
$$= (0.99 \times 0.0001) + (0.005 \times 0.99) = 0.0$$

$$P(D|T) = 0.01943$$

- $P(D) = 0.02$ for $n=20$

$P(D') = 0.98$

(a) For $x=3$,

$$P(x=3) = {}^{20}C_3 \times (0.02)^3 \times (0.98)^{17}$$
$$= 0.0064$$

(b) For at least 2,

lets find for 0, & 1 and negate

$$P(x=0) = {}^{20}C_0 (0.02)^0 \times (0.98)^{20}$$
$$= 0.6676$$

$$P(x=1) = {}^{20}C_1 (0.02)^1 (0.98)^{19}$$
$$= 0.2762$$

$$P(x \geq 2) = 1 - (0.6676 + 0.272)$$
$$= 0.0604$$

- Distribution for 500 hrs.

$$\lambda = \frac{1}{500}$$

for timing, $P(T > t = e^{-\lambda t}$

(a) For b/w 400 & 600 hrs,

$$P(400 \leq T \leq 600) = e^{-400/500} - e^{-600/500}$$

(b) For 250 hrs,

$$P = e^{-250/600}$$

5. $A = \begin{bmatrix} 5 & 2 \\ 5 & 2 \end{bmatrix}$

<u>eigen value</u> $A - \lambda I$

$$\begin{bmatrix} 5-\lambda & 2 \\ 2 & 2-\lambda \end{bmatrix}$$

det $(5-\lambda)(2-\lambda) - 4 = 0$

$10 - 5\lambda - 2\lambda + \lambda^2 - 4 = 0$

$\lambda^2 - 7\lambda + 6 = 0$

$\lambda^2 - \lambda - 6\lambda + 6 = 0$

$\lambda(\lambda - 1) - 6(\lambda - 1) = 0$

$(\lambda - 6)(\lambda - 1) = 0$

$\lambda = 6, 1$

<u>eigen vector</u> $(A - \lambda I)V = 0$

For $\lambda = 6$, $A - 6I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}$

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$-x + 2y = 0$, $2x - 4y = 0$

$2y = x$, $x = 2y$

if $x = 1$ then $y = 2$
$x = 2$ $y = 4$ } every value of x, y doub

(B) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix}$

<u>Eigen value</u> $B - \lambda I = 0$

$$\begin{bmatrix} 1-\lambda & 1 & 0 \\ 0 & 2-\lambda & 0 \\ -1 & 1 & 3-\lambda \end{bmatrix} = 0$$

det $(1-\lambda)(2-\lambda)(3-\lambda) = 0$

$(1-\lambda)(6 - 2\lambda - 3\lambda + \lambda^2) = 0$

$\lambda^2 - 3\lambda - 2\lambda + 6 - \lambda^3 + 3\lambda^2 + 2\lambda^2 - 6\lambda = 0$

$\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

$\lambda^2(\lambda + 6) - 11(\lambda +$

$\lambda = 1, 2, 3$

<u>Eigen vectors</u> $(B - \lambda I)V = 0$

for $\lambda = 1$, $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 2 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$x = 0$, $y = 0$, $z = 0$

for $\lambda = 2$, $\begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ -1 & 1 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$-x + x = 0$, $y \neq 0$; $-z + z + z = 0$

$2z - z = 0$

$\boxed{z = 0}$

for $\lambda = 3$, $\begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 0 \\ -1 & 1 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$-2x + x = 0$ $y = 0$ z not defined